



Buying Tips

- ✓ **BATCH PROCESSING:** Batch processing comes in handy when you have a large number of documents to scan. With this feature you need not sit in front of the scanner feeding one page at a time. Batch processing does need support from hardware as well; you should have a scanner that has an Automatic Document Feeder (ADF).
- ✓ **LANGUAGES:** Check for the languages supported by the OCR package.
- ✓ **LEARNING MODE:** Certain OCR software come with the learning mode feature that allows it to 'learn' from the corrections you make in the recognised document.
- ✓ **SAVE AS:** Apart from saving scanned files as Word documents, look for a Save As option that lets you save them in HTML, PDF and EXCEL formats as well.
- ✓ **FONT TRAINING:** The OCR software should have a provision to add specific fonts and letter types into its dictionary, in case it fails to pick them up while scanning your documents.

Quick Find

YOU NEED	A cost-effective solution for home use
Basic Use	
LOOK FOR	An OCRing software that supports documents containing text and tables with basic formatting, the occasional newspaper and magazine clipping, with average accuracy
YOU NEED	A high-end software for professional usage
Inter Use	
LOOK FOR	A software that OCRs faxes, photocopies, type-written documents, newspapers, magazines, contracts, and tenders containing text and tables, with high levels of accuracy

Why OCR?

The benefits of OCR become clearer after you start using it. Though it's generally considered a tool for enterprises, home-users and small offices can also benefit from it. To start with, you can scan all your business cards, OCR them and maintain a searchable archive of them—you can even use special software to import scanned business cards.

Utilities called form fillers enable you to scan printed forms and fill them in electronically. Storing these forms electronically prevents the need to rewrite the entire form each time a mistake is made. Also, you can fill in multiple forms simultaneously—a feature that comes in handy for railway reservation forms.

How does OCR work?

OCR is used to scan and recognise text on paper, or convert scanned pages saved as images to text. It analyses the raster image and creates an index of areas that resemble possible text fields. The software then attempts to recognise text characters by comparing the shape of the scanned objects to a database of words categorised by different fonts or typefaces.

Thereafter, it groups the individual characters and compares them with the words in the dictionary that is set to use a particular language. This step is extremely crucial for recognition accuracy. The more comprehensive the dictionary, the more

accurate is the finished product.

The OCR software marks certain words that it 'considers' inaccurate for you to correct them manually.

Finally, the OCR software uses the index it created to align the text fields as accurately as possible.

The accuracy of the OCR software depends on many factors, ranging from scanner quality to the quality of the paper on which the text is printed.

The latest breed of OCR packages uses optimisation algorithms, neural networks and even AI concepts to get this done. Using pattern recognition techniques, these software try to guess the character as a whole and look at all possibilities before arriving at a hypothesis. For example, it may guess that a character is an 'H' even if the scanned document doesn't show a horizontal bar in the image. Some OCR packages have inbuilt tools that enable them to 'learn' from the changes you make to the output.

Factors that affect OCR software accuracy

- Scanner quality
- Scan resolution
- Type of printed documents, whether laser printer outputs or photocopied
- Paper quality
- Fonts used in the text
- Linguistic complexities
- The dictionary used

Do Remember

- A document that is free of wrinkles, tears, smudges or creases yields more accurate results.
- Keep a sheet of white paper behind a document while OCRing it, especially when scanning newspapers, thin paper, etc. This helps eliminating smudges during scanning.

The Future

From just recognising standard fonts to predicting the word itself and now 'correcting' broken words, OCR software has improved by leaps and bounds. Learning OCR is already up and running with inbuilt tools that help enhance self-learning ability. Symbol recognition is another aspect of OCR that is increasingly becoming

ing a regular feature of software packages. Field-specific recognition, wherein the scanned data is automatically stored in the appropriate field in, say, a database, will also be an integral part of the software in the near future.

